Title of Project Report

Author Names

Group Number

Date: yyyy-mm-dd

Abstract

Briefly summarize the problem and its importance, the method used, the main results, and the significance of those results.

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1 Introduction

1.1 Background

Introduce the relevant background and explain why the problem is worth studying.

1.2 Problem Statement

Briefly describe the problem, the purpose of this report, and any relevant background on the data structures and algorithms.

1.3 Objectives

State the goals of the project and summarize the expected outcome.

2 Data Structure / Algorithm Specification

2.1 Data Structures

Describe the main data structures, their stored information, and the operations required by the solution.

2.2 Algorithm Design

Describe all algorithms used to solve the problem (pseudocode is preferred), and specify the main data structures. Cite adopted algorithms properly, for example the standard algorithms textbook [1].

2.3 Correctness

Explain why the algorithm produces the required result. Add lemmas or a formal proof when appropriate.

3 Testing Results

3.1 Environment and Methodology

Describe the hardware, software, datasets, and testing procedure.

3.2 Test Cases

For each test, record the concrete input or scenario under *Test Case*, and classify its role under *Test Type*, such as a functional test, a medium-scale data test, or a large-scale data test. Also record the expected result, actual result, and status. For a failed test, explain the possible cause and correction in the discussion below the table.

表 1: Testing results

ID	Test Case	Test Type	Expected Result	Actual Result	Status
1	Small hand-checkable input	Functional test	Correct output	Same as expected	Pass
2	10^4 randomly generated records	Medium-scale data test	Correct output within the target time	Record the measured result	Pass
3	10^6 randomly generated records	Large-scale data test	Correct output without failure	Record the measured result	Pass

3.3 Results and Discussion

Summarize what the tests demonstrate and discuss unexpected behavior or limitations.

4 Analysis and Comments

4.1 Time Complexity

Analyze the running time of the main operations and the complete algorithm.

4.2 Space Complexity

Analyze auxiliary storage and total memory usage.

4.3 Comparison and Improvements

Compare the solution with other known data structures or algorithms, discuss its trade-offs, and propose possible improvements.

Reference

[1] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, 3rd ed. Cambridge, MA, USA: The MIT Press, 2009.

5 Author List

5.1 Division of Work

Specify each member's contribution to justify the author list on the cover.

- Author: Code, experiments, and report.
- Reviewer: Reproduction, review, and feedback.

5.2 Collaboration Summary

Briefly describe how the group integrated, reviewed, and verified its work.

6 Declaration

We hereby declare that all work in this project, titled "Title of Project Report," is our independent effort as a group.

7 Signatures

Author Names

8 Source Code Appendix

The full source code is provided in the accompanying `code/` directory.